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Homework 9

**1.**

**Exercise 4.9** Show that

$$\int_{-\infty}^{\infty} \exp\{-x^2\} \exp\{-(x-\mu)^2 y/2\sigma^2\} \, \mathrm{d}x = \exp\left\{-\frac{\mu^2}{1+2\sigma^2/y}\right\}$$

by completing the square in the exponent to evaluate the integral.

We start with

$$\begin{aligned}
&\int_{-\infty}^{\infty} \exp\{-x^2\} \exp\{\frac{-(x-\mu)^2 y}{2\sigma^2}\} dx \\
&= \int_{-\infty}^{\infty} \exp\{-x^2 - \frac{(x-\mu)^2 y}{2\sigma^2}\} dx \\
&= \int_{-\infty}^{\infty} \exp\{-x^2 - \frac{(x^2+\mu^2-2x\mu)y}{2\sigma^2}\} dx \\
&= \int_{-\infty}^{\infty} \exp\{-x^2 - \frac{x^2 y}{2\sigma^2} - \frac{y\mu^2}{2\sigma^2} + \frac{xy\mu}{\sigma^2}\} dx \\
&= \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})x^2 + \frac{xy\mu}{\sigma^2} - \frac{y\mu^2}{2\sigma^2}\} dx
\end{aligned}$$

With this expression equating to the form of $-ax^2 + bx + c$, we can try to complete the square as instructed, which would yield $-a(x - \frac{b}{2a})^2 + \frac{b^2}{4a} + c$. Let's try to match the format with $a = 1 + \frac{y}{2\sigma^2}$, $b = \frac{y\mu}{\sigma^2}$, and $c = -\frac{y\mu^2}{2\sigma^2}$:

$$\begin{aligned}
\frac{b}{2a} &= \frac{\frac{y\mu}{\sigma^2}}{2 + \frac{y}{\sigma^2}} \\
&= \frac{\frac{y\mu}{\sigma^2}}{\frac{2\sigma^2+y}{\sigma^2}} \\
&= \frac{y\mu}{2\sigma^2 + y}
\end{aligned}$$

and

$$\begin{aligned}
\frac{b^2}{4a} &= \frac{\frac{y^2\mu^2}{\sigma^4}}{4 + \frac{2y}{\sigma^2}} \\
&= \frac{\frac{y^2\mu^2}{\sigma^4}}{\frac{4\sigma^2+2y}{\sigma^2}}
\end{aligned}$$

$$= \frac{y^2\mu^2}{4\sigma^4 + 2y\sigma^2}$$

So plugging in:

$$\int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x - \frac{y\mu}{2\sigma^2+y})^2 + \frac{y^2\mu^2}{4\sigma^4+2y\sigma^2} - \frac{y\mu^2}{2\sigma^2}\}dx$$

$$= \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x - \frac{y\mu}{2\sigma^2+y})^2 + \frac{y^2\mu^2 - y\mu^2(2\sigma^2+y)}{2\sigma^2(2\sigma^2+y)}\}dx$$

$$= \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x - \frac{y\mu}{2\sigma^2+y})^2 + \frac{y^2\mu^2 - 2y\mu^2\sigma^2 - y^2\mu^2}{2\sigma^2(2\sigma^2+y)}\}dx$$

$$= \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x - \frac{y\mu}{2\sigma^2+y})^2 + \frac{-2y\mu^2\sigma^2}{2\sigma^2(2\sigma^2+y)}\}dx$$

$$= \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x - \frac{y\mu}{2\sigma^2+y})^2 + \frac{-y\mu^2}{2\sigma^2+y}\}dx$$

$$= \exp\{\frac{-y\mu^2}{2\sigma^2+y}\} \int_{-\infty}^{\infty} \exp\{-(1+\frac{y}{2\sigma^2})(x - \frac{y\mu}{2\sigma^2+y})^2\}dx$$

The integral is the Gaussian integral, with the form $\int_{-\infty}^{\infty} e^{-ax^2}dx = \sqrt{\frac{\pi}{a}}$. Using this:

$$= \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}\sqrt{\frac{\pi}{1+\frac{y}{2\sigma^2}}}$$

$$= \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}\sqrt{\frac{\pi}{\frac{2\sigma^2+y}{2\sigma^2}}}$$

$$= \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}\sqrt{\frac{2\pi\sigma^2}{2\sigma^2+y}}$$

With the additional term in the corrected version of the problem:

$$= \frac{1}{\sqrt{\frac{2\pi\sigma^2}{y}}} \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}\sqrt{\frac{2\pi\sigma^2}{2\sigma^2+y}}$$

$$= \sqrt{\frac{y}{2\pi\sigma^2}} \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}\sqrt{\frac{2\pi\sigma^2}{2\sigma^2+y}}$$

$$= \sqrt{\frac{y}{2\pi\sigma^2} \cdot \frac{2\pi\sigma^2}{2\sigma^2+y}} \exp\{\frac{-y\mu^2}{2\sigma^2+y}\}$$

$$
\begin{aligned}
&= \sqrt{\frac{y}{2\sigma^2+y}}\exp\{\frac{-y\mu^2}{2\sigma^2+y}\} \\
&= \sqrt{\frac{y}{2\sigma^2+y}}\exp\{\frac{-y\mu^2}{2\sigma^2+y}\} \\
&= \frac{1}{\sqrt{\frac{2\sigma^2+y}{y}}}\exp\{\frac{-y\mu^2}{2\sigma^2+y}\} \\
&= \frac{1}{\sqrt{\frac{2\sigma^2}{y}+1}}\exp\{\frac{-y\mu^2}{2\sigma^2+y}\} \\
&= \frac{1}{\sqrt{\frac{2\sigma^2}{y}+1}}\exp\{\frac{-y\mu^2}{2\sigma^2+y}\} \\
&= \frac{1}{\sqrt{\frac{2\sigma^2}{y}+1}}\exp\{-\frac{\mu^2}{\frac{2\sigma^2+y}{y}}\} \\
&= \boxed{\frac{1}{\sqrt{\frac{2\sigma^2}{y}+1}}\exp\{-\frac{\mu^2}{1+\frac{2\sigma^2}{y}}\}}
\end{aligned}
$$

exactly as we sought to show in the corrected version of the problem.

**2.**

**Exercise 4.10** Consider an Accept–Reject method for the target density $f$ based on the instrumental distribution $g$.

a. Show that the Accept–Reject sample $(X_1, \ldots, X_n)$ can be associated with two iid samples, $(U_1, \ldots, U_N)$ and $(Y_1, \ldots, Y_N)$, with corresponding distributions $\mathcal{U}_{[0,1]}$ and $g$; $N$ is the stopping time associated with the acceptance of $n$ variables $Y_j$.

b. Deduce that the corresponding estimator of $\mathbb{E}_f[h(X)]$ based on $(X_1, \ldots, X_n)$ can be written as

$$\delta_1 = \frac{1}{n}\sum_{i=1}^{n} h(X_i) = \frac{1}{n}\sum_{j=1}^{N} h(Y_j)\,\mathbb{I}_{U_j \le w_j}\,,$$

with $w_j = f(Y_j)/Mg(Y_j)$.

**(a)**

**(b)**

**3.**

**Exercise 4.15** For each of the following cases, generate random variables $X_i$ and $Y_i$ and compare the empirical average and Rao–Blackwellized estimator of $\mathbb{E}_f(X)$ and $\mathrm{var}_f(X)$:

a. $X|y \sim \mathcal{P}(y)$, $Y \sim \mathcal{G}a(a,b)$ ($X$ is negative binomial);
b. $X|y \sim \mathcal{N}(0,y)$, $Y \sim \mathcal{G}a(a,b)$ ($X$ is a generalized $t$);
c. $X|y \sim \mathcal{B}in(y)$, $Y \sim \mathcal{B}e(a,b)$ ($X$ is beta-binomial).

**(b)**

**(c)**

**4.**

**Exercise 4.18** A naïve way to implement the antithetic variable scheme is to use both $U$ and $(1-U)$ in an inversion simulation. Examine empirically whether this method leads to variance reduction for the distributions (i) $f_1(x) = 1/\pi(1+x^2)$, (ii) $f_2(x) = \frac{1}{2}e^{-|x|}$, (iii) $f_3(x) = e^{-x}\mathbb{I}_{x>0}$, (iv) $f_4(x) = \frac{2}{\pi\sqrt{3}}\left(1+x^2/3\right)^{-2}$, and (v) $f_5(x) = 2x^{-3}\mathbb{I}_{x>1}$. Examine variance reductions of the mean, second moment, median, and 75th percentile.

To calculate the weights for the Rao–Blackwellized estimator of Section 4.6, it is necessary to derive properties of the distribution of the random variables in the Accept–Reject algorithm. The following problem is a rather straightforward exercise in distribution theory and is only made complicated by the stopping rule of the Accept–Reject algorithm.

**(i)**

**(ii)**

**(iv)**

**5.**

Assuming that we use antithetic variables to estimate some parameter of the standard normal distribution, we must prove that the covariance between $X_i$ and $Y_i$ is -1, i.e. $\mathrm{Cov}(X_i, Y_i) = -1$.

We set up with $U \sim \mathrm{Unif}(0,1)$. Using the hint with the normal distribution be-

ing symmetric about 0, we know that $\Phi(z) = 1 - \Phi(-z)$ for the range of $U$, and equivalently that

$$x = \Phi(z) = 1 - \Phi(-z)$$

$$\Phi(-z) = 1 - x$$

$$-z = \Phi^{-1}(1 - x)$$

$$-\Phi^{-1}(-x) = \Phi^{-1}(1 - x)$$

So taking advantage of this, let's set $X = \Phi^{-1}(U)$ and $Y = \Phi^{-1}(1 - U)$ accordingly. We can then determine that

$$Y = \Phi^{-1}(1 - U) = -\Phi^{-1}(-U) = -X$$

So therefore

$$\begin{aligned} \mathrm{Cov}(X_i, Y_i) &= \mathrm{Cov}(X_i, -X_i) = -\mathrm{Cov}(X_i, X_i) \\ &= -\mathrm{Var}(X_i) = \boxed{-1} \end{aligned}$$

just as we aimed to prove.